Reviewer Pack — Invariants of Quasi-Symmetric Functions and Communication Co…

Entry db9454c867d7 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-27 23:08:41 UTC

Invariants of Quasi-Symmetric Functions and Communication Complexity

Entry ID: `db9454c867d7` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-27 23:08:41 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics **Field B** (complexity object): Communication Complexity

Statement:

> For any Boolean function f, the communication complexity CC(f) is upper-bounded by a polynomial in the number of quasi-symmetric functions φ such that φ^2 = f, where φ is a homomorphism from the free group on two generators to the symmetric group on n elements.

Rationale (proposer's reasoning):

> Quasi-symmetry in algebraic combinatorics has been used to analyze structures with symmetry properties. If this property can be linked to communication complexity, it could provide new insights into the lower bounds of communication protocols. The polynomial upper bound ensures that the conjecture is testable within the given computational constraints.

Taxonomy category: `GCT_DET_PERM` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `dd85b80b5cf7dfde`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The communication complexity CC(f) for Boolean function f is considered supported if it is upper-bounded by a polynomial of degree D in the number of quasi-symmetric functions φ such that φ^2 = f, and falsified otherwise.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `"quasi-symmetric functions" AND "communication complexity"` - `"polynomial upper bound" IN algcomb AND IN commcomplex` - `"homomorphism" AND free group AND symmetric group AND Boolean function`

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=-9, elapsed=65.0s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def homomorphism_count(f, n):
        count = 0
        G = list(range(2))
        S_n = list(itertools.permutations(range(n)))
        for phi in itertools.product(S_n, repeat=4):
            if all(phi[phi[g][g]] == f[g] for g in G):
                count += 1
        return count

    def communication_complexity(f, n):
        # Placeholder function for actual CC computation
        return len(f) ** 0.5  # Simplified example

    instances_tested = 0
    total_metric_value = 0
    conjecture_holds = True
    counterexample = ""

    for _ in range(30):  # Ensure at least 30 instances per seed
        n = random.choice([5, 10, 15, 20, 30, 40])
        f = generate_boolean_function(n)
        phi_count = homomorphism_count(f, n)
        cc = communication_complexity(f, n)
```

```
        if phi_count == 0:
            conjecture_holds = False
            counterexample = "phi_count_zero"
            break

        instances_tested += 1
        total_metric_value += cc

    if instances_tested == 0:
        return {
            "metric_name": "communication_complexity",
            "metric_value": 0,
            "instances_tested": 0,
            "conjecture_holds": False,
            "counterexample": "no_instances"
        }

    mean_metric_value = total_metric_value / instances_tested
    return {
        "metric_name": "communication_complexity",
        "metric_value": mean_metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 103, 4))  # Default to first 30 primes

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0 support_fraction={support_fractio
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"phi_count_zero\" first_failing_seed={first_fail
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
(empty)
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test crashed before producing data, which means the pre-registered support condition could not be unambiguously met. | next: Re-run the test to ensure it completes successfully and provides results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10391
2	preregistration	ollama_remote	glm4:latest	0	0	5650
3	novelty	ollama_remote	glm4:latest	0	0	4597
4	novelty	ollama_remote	glm4:latest	0	0	5263
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12725
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9404
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8271
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8561
9	judge	ollama_remote	glm4:latest	0	0	58511

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 123371 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/db9454c867d7.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/db9454c867d7.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/db9454c867d7.tar.gz` (if generated)